Combinations

1

a 120 **b** 17280 **c** 462 **d** 3876

e 5 **f** 1260 **g** 21 **h** 1

i 1 **j** 35 **k** 10 **l** $\frac{6}{5}$

2

a $(n+1)!$ **b** $n+1$ **c** $n(n-1)$ **d** $\frac{1}{(n+5)(n+6)}$

3

a $n=7$ **b** $n=3,4$

4

a No solution provided **b** No solution provided

c No solution provided **d** No solution provided

e No solution provided **f** No solution provided

Applications

5

a

	C	P	W
B	C,B	P,B	W,B
G	C,G	P,G	W,G
R	C,R	P,R	W,R
Y	C,Y	P,Y	W,Y

b 12

6 924

7 $20\,475$

8 $3\,108\,105$

9 2925

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11 126

12 42 504

13 35

14 128

15
a 1820 **b** 66

16 125 970

17 91

18 6006

19 7350

20 27 720

21
a 4512 **b** 17 296 **c** 21 808

22 211 926

23
a 252 **b** 210 **c** 126

24 128

25 16 777 216

26
a 15 **b** 63

27 20

28
a 165 **b** 95

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29 120

30
a 20 **b** 6 **c** 30

31
a 150 **b** 310

32 30

33 126

34
a 5 **b** 9 **c** ${}^{n}C_{2} - n$

35 76

36 672

37 435

38
a 1540 **b** 120 **c** $\frac{6}{77}$

39
a 475 020 **b** 12 376 **c** $\frac{34}{1305}$

40
a 10 626 **b** 5 **c** $\frac{5}{10\,626}$

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